

ALVINE MUGO MUNGAI

Full Stack Developer (Backend-Focused) Nairobi, Kenya | +254 716 586 275 | alvnmugo@gmail.com | GitHub: github.com/alvn2 | LinkedIn: linkedin.com/in/alvine-mugo | Portfolio: alvinemugo.vercel.app

Core Technologies: Node.js · Express.js · Flask · PostgreSQL · MongoDB · React · Next.js · TypeScript · REST APIs · Docker · JWT · RBAC · Git

PROFESSIONAL SUMMARY

Software Engineer and Technology Consultant with a track record of architecting scalable, production-ready digital infrastructure. Proficient in Next.js, React, and Node.js on the frontend, with a strong backend specialization in Express, Flask, PostgreSQL, and MongoDB. Experienced in privacy-first design and building secure, high-performance systems for NGOs, SMEs, and enterprise clients across Kenya. Focused on translating complex business requirements into clean, tested, and maintainable software that drives real operational impact.

TECHNICAL SKILLS

Category	Skills
Backend	Node.js, Express.js, Flask (Python), RESTful API Design, JWT Authentication, RBAC, Middleware, MVC Architecture
Databases	PostgreSQL, MongoDB, MySQL, Redis, Schema Design, Query Optimization, Indexing, NoSQL Modeling
Frontend	React.js, Next.js, TypeScript, JavaScript (ES6+), Tailwind CSS, Redux Toolkit, Context API, Chart.js, Recharts
DevOps & Tools	Git, GitHub, Docker, Linux, Vercel, Firebase, Supabase, Google Cloud Platform (GCP), CI/CD, Postman, PM2
Testing	Jest, Minitests (Python), Test-Driven Development (TDD), 85%+ test coverage
Concepts	Agile, System Architecture, OOP, Data Structures & Algorithms, Privacy-First Design, Client-Side Encryption

PROFESSIONAL EXPERIENCE

Full Stack Developer (Volunteer) | Friends of the Youth (FOTY) Nairobi, Kenya | May 2024 – Present

- Architected and deployed a custom SPA (foty.co.ke) serving a national youth NGO with thousands of members across Kenya, replacing legacy systems with a React/Next.js frontend
- Engineered automated data collection workflows improving donor reporting accuracy and operational efficiency
- Implemented role-based security protocols for sensitive beneficiary data and optimized database queries for real-time reporting
- Provide ongoing maintenance, feature enhancements, and technical support ensuring platform reliability

Software Engineer & Tech Consultant | DevLink Technologies Nairobi, Kenya | November 2024 – Present

- Design and develop full-stack web applications for clients across e-commerce, agriculture, and education sectors using Next.js, React, Node.js, and Flask/PostgreSQL
- Build and document RESTful APIs with proper authentication, input validation, and error handling
- Deliver 5+ production-ready applications with 100% client satisfaction, improving average client operational efficiency by 40%
- Reduced application downtime by 20% through proactive monitoring and performance optimization
- Built "BizScout," a lead generation tool automating Google Maps data extraction for local service providers
- Architected high-performance inventory catalog solutions for retail clients, reducing customer inquiry time by 60% via automated WhatsApp ordering workflows
- Developing "Omniflow," a cross-platform media player featuring mood-based algorithms and custom audio rendering

Software Testing & Debugging Intern | AlgoFed Nairobi, Kenya | November 2024 – February 2025

- Conducted manual and automated testing identifying 50+ bugs across multiple projects, contributing to a 15% reduction in production errors
 - Wrote and executed test cases covering edge cases, user workflows, and error handling
 - Improved application performance by 30% through bottleneck identification and query optimization
 - Collaborated with senior engineers to prioritize and resolve critical issues
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KEY PROJECTS

ShopOS — Auto Parts Inventory Management System *(Fully Deployed) React 19, TypeScript, Vite, Node.js, Express, Google Sheets API (GCP), Tailwind CSS, Recharts*

- Architected a full enterprise ERP system for auto parts retail, replacing expensive proprietary software (Oracle/SAP) for SMEs
- Engineered a Node.js/Express RESTful API integrating Google Sheets as a cost-effective, highly accessible database layer via GCP service accounts — handling caching, batch writes, and retry logic to manage API rate limits
- Implemented a complete RBAC system with an immutable AuditLog recording every destructive action (create, update, delete) with user UUID, timestamp, and action context
- Built optimistic UI updates in React for POS operators, providing instant feedback while reconciling with the backend asynchronously
- Features include real-time inventory tracking, bulk point-of-sale processing, dynamic financial reporting with Recharts, admin session management, and a database-driven settings system

FarmLink — Agricultural Management Platform *(Deployed) React, Node.js, Express, MongoDB, JWT, Chart.js, Tailwind CSS | [GitHub](#)*

- Architected complete MERN stack application from requirements to deployment for Kenyan smallholder farmers
- Built secure RESTful API with JWT authentication and role-based authorization
- Designed MongoDB schema with efficient data relationships across farms, crops, and tasks
- Delivered real-time data visualization dashboards using Chart.js; streamlined farm productivity by 35%

SafeHaven — Mental Health Platform *(In Active Development)* *Next.js, Supabase, AES-GCM, Argon2id*

- Building a zero-knowledge, privacy-first platform for anonymous mental health support in Kenya
- Implementing client-side AES-GCM + Argon2id encryption to ensure absolute user anonymity and data sovereignty
- Engineered a "Quick Exit" panic button and encrypted local-first data storage for at-risk users

Asset Inventory Management System *(Frontend Deployed — Backend in Staging)* *Flask, PostgreSQL, React, Redux Toolkit, Jest, Minitests* | [GitHub](#)

- Developed enterprise-level asset tracking system with three-tier RBAC (Admin, Manager, User)
 - Built RESTful API with Flask following MVC architecture; designed and optimized PostgreSQL schema
 - Achieved 85%+ test coverage using Jest (frontend) and Minitests (backend)
 - Reduced asset mismanagement risk by 40% through automated audit trails and real-time status tracking
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EDUCATION

Software Engineering Scholarship | Power Learn Project (PLP) — Remote February 2025 - December 2025 Specialization in MERN Stack Development, Cloud Deployment, and Microservices Architecture. Delivered production-grade capstone project.

Full Stack Web Development | Moringa School — Nairobi, Kenya May 2024 - November 2024 Intensive 6-month bootcamp. Built 10+ full-stack applications. Mastered React, Node.js, Express, MongoDB, and Agile development practices.

CERTIFICATIONS

- Microsoft Azure AI Essentials — July 2025
 - Mastering MySQL: Database Creation, Management & SQL Queries — Udemy, November 2024
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